

c) Every student in this class has taken at least one computer science course

$T(x, y, z)$: Person x has taken course y from department z .

Domain of y : Set of all courses in the school

Domain of z : Set of all departments in the school

$\forall x \exists y T(x, y, \text{Computer Science})$

d) There is a student in this class who has taken at least one course in computer science

$\exists x \exists y T(x, y, \text{Computer Science})$

e) Every student in this class has been in every building on campus.

$V(x, y)$: Student x has visited building y

Domain of y : Set of all buildings on campus

$\forall x \forall y V(x, y)$

f) There is a student in this class who has been in every room of at least one building on campus.

$V(x, y, z)$: Person x has visited room y of building z .

Domain of y : Set of all rooms on campus

Domain of z : Set of all buildings on campus

$\exists x \exists z \forall y V(x, y, z)$

g) Every student in this class has been in at least one room of every building on campus

$\forall x \forall z \exists y V(x, y, z)$

16) A discrete mathematics class contains 1 mathematics major who is a freshman, 12 mathematics majors who are sophomores, 15 computer science majors who are sophomores, 2 mathematics majors who are juniors, 2 computer science majors who are juniors, and 1 computer science major who is a senior. Express each of these statements in terms of quantifiers and then determine its truth value.

$M(x)$: x is a Mathematics major $C(x)$: x is a computer science major